

Research in Astrophysics from Space (E)
Solar jets, CMEs, and Associated EUV Waves (E2.4)
Either poster or oral presentation (no preference).

THE SEARCH FOR SOLAR JETTING IN PARKER WISPR DATA IN ORDER TO DETERMINE THE EXTENTS OF THEIR TOTAL COMPOSITION OF THE SOLAR WIND

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Small jets are nearly ubiquitous across the solar surface and occur regularly over the Sun and throughout the solar cycle, and one interesting idea from several recent observational studies is that there are enough small jets to energize and fill most of the solar wind [Raouafi et al. 2023 doi: 10.3847/1538-4357/acaf6c; Chitta et al., 2023, doi: 10.1126/science.ade5801]. The largest solar jet ejecta viewed in the Solar Dynamics Observatory/Atmospheric Imaging Assembly (SDO/AIA) are possible to distinguish individually and can be traced outward from the solar surface into the heliosphere [Yu et al., 2016, doi: 10.1002/2016JA022503]. The X-Ray bright flashes associated with solar jetting have been observed in the Hinode X-Ray Telescope and used to determine the energy of many events. These values were used to extrapolate from the highest down to the lowest intensity jets and integrate the contribution to the solar wind energy and associated mass from all the jets. This indicated that the smallest jets are insufficient to provide a large fraction of the solar wind outflow for either energy or mass [Yu et al., 2014, doi: 10.1088/0004-637X/784/2/166].

However, this question is still an open one with different kinds of small-scale jetting now known from more examples such as the previously mentioned small jet observations as well as “inconspicuous jets” which were found first from jet-like Doppler outflows and then matched to faint-but-confirmed jet events in AIA images [Sterling et al., 2022, doi: 10.3847/1538-4357/ac9960]. Previous statistical studies have found matching levels of activity between solar jets and switchbacks in the solar wind as seen by the Parker Magnetometer [Bizien et al., 2025,

doi: 10.1051/0004-6361/202452140] but a direct causal correlation between jets on the solar surface and dense patches in the solar wind has not yet been achieved. Far better tools are now at our disposal to learn jetting ejected mass and energies from Parker Solar Probe WISPR data. In this ongoing study we can now determine these characteristics of the largest of the jetting events using time-dependent 3-D reconstructions [Jackson et al., 2020, doi: 10.3389/fspas.2020.568429] of Parker data to within three solar radii of the solar surface. This allows for better analysis of the largest events, which when extrapolated to zero as in Yu et al., [2014, doi: 10.1088/0004-637X/784/2/166], now give a more precise estimate of the extent to which jetting events can completely populate and extend outward into the solar wind to provide its total energy and mass as mesoscale structure [Viall et al., 2021, doi: 10.3389/fspas.2021.735034].